

Chapter 4

EFFECTS OF ATMOSPHERIC PARTICULATE MATTER ON MATERIALS

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Chapter 4

EFFECTS OF ATMOSPHERIC PARTICULATE MATTER ON MATERIALS

A. INTRODUCTION

Airborne particles may damage surfaces merely by settling on them. A light deposit of dust makes surfaces appear dingy, and the frequent cleaning necessary in a dusty atmosphere weakens materials and costs money.

Particles can also cause direct or chemical deterioration of materials. The nature and extent of the deterioration depends on the chemical activity of the particles in their environment and the relative susceptibility of the receiving material. Particles may cause chemical deterioration either by acting as condensation nuclei for the retention of adsorbed gases or harmful acids, or by their own innate corrosive action.

In the following sections it will be seen that particulate air pollution plays an important role in the corrosion of metals; in the soiling, damage, and erosion of coatings and painted surfaces, building stones, marble, and other building materials; and in the soiling and degradation of textiles.

B. EFFECTS OF PARTICULATE MATTER ON METALS

Atmospheric particles may accelerate the corrosion of iron, steel, and various nonferrous metals.

Metals are generally resistant to attack in dry air,¹ and even clean moist air does not cause significant corrosion.^{1, 2} Furthermore, inert dust and soot particles without sulfur compounds as constituents do not of themselves cause marked corrosion.^{1, 3} Particles may, however, contribute to accelerated corrosion in two ways.⁴ First, they may be intrinsically active, and secondly, although inactive, they may be capable of absorbing or adsorbing active gases (such as SO_2) from the atmosphere.

Those particles which are inactive and have negligible capacity for absorption or adsorption have little effect other than that of acting as droplet nuclei in the atmosphere. For example, silica particles do not accelerate the rate of metal corrosion even in the presence of SO_2 . On the other hand, charcoal (carbonaceous) particles in atmospheres with relative humidities below 100 percent cause a large increase in the rate of corrosion in the presence of SO_2 traces, presumably through the local concentration of the gas by adsorption.⁴ The laboratory research which led to these findings was based on particulate concentrations of 0.4 mg/cm^2 (equivalent to 0.3 tons/mi^2) and abnormally high SO_2 concentrations of 100 ppm.

Active hygroscopic particles such as sulfate and chloride salts and sulfuric acid aerosol serve as corrosion nuclei. Their presence in the atmosphere can initiate corrosion, even at low relative humidities.⁵ Laboratory studies of bare and varnished steel test panels, properly polished and degreased, and then inoculated with finely divided particles of various substances such as are commonly found in the atmosphere, were conducted by Preston and Sanyal.⁵ Particles of chloride, sulfate, chromate, and oxide salts, and boiler and flue dusts were used. The metal test panels were exposed to atmospheres of pure clean air and of air containing SO_2 at various humidities, and the resulting corrosion was measured. Filiform corrosion, characterized by a filamental configuration, the primary phase in electrolytic corrosion, was noted in all cases. Corrosion rates are low when the relative humidity is below 70 percent, but they increase at higher humidities.^{4, 6} In most of the cases in this study, corrosion increased with increased humidity even in clean air. The addition of traces of

SO₂ to the test atmosphere greatly increased the rate of corrosion in all instances.⁵

Field experiments show that the rate of corrosion of various metals is accelerated in urban and industrial areas because of the greater atmospheric concentrations of both particulate matter and sulfur compounds. Standardized open-hearth iron specimens, after exposure for one year at a number of diverse locations throughout the world, were observed by Hudson⁷ to have a manifold variation in weight loss. Iron specimens exposed for one year to desert and arctic environments, the least polluted areas, were the least corroded, i.e., had the smallest losses in weight. Those specimens exposed at heavily-polluted urban industrial sites were the most corroded, more so than similar specimens exposed at many marine and tropical locations, Table 4-1. Hudson, Figure 4-1, also correlated the rate of rusting of mild steel specimens and dustfall levels for four diverse areas, from a heavily industrialized area in Sheffield, England, to a rural area (Llanwrtyd Wells). The iron specimens corroded four times as fast in the polluted industrial atmosphere as they did in the rural atmosphere.

Committee B-3 of the American Society of Testing Materials (ASTM)^{8,9} studied cor-

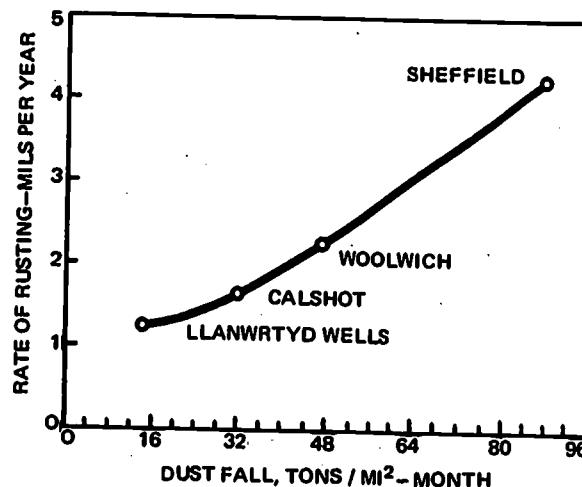


FIGURE 4-1. Plot of Rate of Rusting Versus Dust-fall at Four Locations.⁷ (The figure plots the rate of rusting of mild steel versus dustfall, and shows that corrosion is four times as rapid in an industrial area (Sheffield) as it is in the rural area (Llanwrtyd Wells)).

rosion rates of steel and zinc panels exposed for one year at several locations in the United States. The relative corrosivities of various atmospheres at 19 sites were compared to that of the rural site of State College, Pennsylvania, which was used as a base. This study (Table 4-2) confirms Hudson's observations that industrial locations with

Table 4-1.--CORROSION OF OPEN HEARTH IRON SPECIMENS IN DIFFERENT LOCATIONS.⁷

Location	Type of atmosphere	Annual weight loss, g	Relative corrosivity
Khartoum, Sudan.....	Dry inland.....	0.16	1
Abisko, North Sweden.....	Unpolluted.....	0.46	3
Aro, Nigeria.....	Tropical inland.....	1.19	8
Singapore, Malaya.....	Tropical marine.....	1.36	9
Basrah, Iran.....	Dry inland.....	1.39	9
Apapa, Nigeria.....	Tropical marine.....	2.29	15
State College, Pa., USA.....	Rural.....	3.75	25
Berlin, Germany.....	Semi-industrial.....	4.71	32
Llanwrtyd Wells, British Isles.....	Semi-marine.....	5.23	35
Calshot, British Isles.....	Marine.....	6.10	41
Sandy Hook, N.J., USA.....	Marine-semi-industrial.....	7.34	50
Congella, South Africa.....	Marine.....	7.34	50
Motherwell, British Isles.....	Industrial.....	8.17	55
Woolwich, British Isles.....	Industrial.....	8.91	60
Pittsburgh, Pa., USA.....	Industrial.....	9.65	65
Sheffield Univ., British Isles.....	Industrial.....	11.53	78
Derby South End, British Isles.....	Industrial.....	12.05	81
Derby North End, British Isles.....	Industrial.....	12.52	84
Frodingham, British Isles.....	Industrial.....	14.81	100

high concentrations of particles and of oxides of sulfur are more corrosive to steel and zinc than less industrialized areas. As shown in Table 4-2, steel specimens corroded in one year approximately 3.1 times as much in New York City (spring exposure) and 3.3 times as much in Kearny, New Jersey, as similar specimens at State College, Pennsylvania. Zinc specimens corroded 3.6 and 2.6 times as much in New York City (spring exposure) and Kearny respectively, as similar specimens exposed at State College. Both steel and zinc specimens in New York City (fall exposure when the particle and SO₂

concentrations in the atmosphere are greater than in spring) corroded 6.0 and 3.7 times as much respectively as similar steel and zinc specimens at State College.^a The authors of the resulting papers did not give mean concentrations of suspended particulate matter for the various locations; but approximate concentrations, based on NASN measurements¹⁰ made after the corrosion studies, are given here. Based on Table 4-3, a town the size of State College, Pa., could have a mean suspended particulate-matter level of 60 µg/m³ to 65 µg/m³. The mean suspended particulate matter concentrations were 176

Table 4-2.—CORROSIVITY OF ATMOSPHERES TOWARDS STEEL AND ZINC AT SELECTED LOCATIONS RELATIVE TO THAT AT STATE COLLEGE, PENNSYLVANIA.^a

Location	Type of atmosphere	Relative corrosivity of one-year test for	
		Steel	Zinc
Norman Wells, N.W.T., Canada	Polar-Rural	0.03	0.4
Esquimalt, Vancouver Is., Canada	Rural-Marine ^a	0.5	0.4
Saskatoon, Sask., Canada	Rural	0.6	0.5
Perrine, Fla.	Rural	0.9	1.0
State College, Pa.	Rural	1.0	1.0
Ottawa, Canada	Semi-Rural	1.0	1.2
Middletown, Ohio	Semi-Industrial	1.2	.9
Trail, B.C., Canada	Semi-Rural	1.4	1.6
Montreal, Que., Canada	Industrial	1.5	2.2
Halifax, N.S., Canada	Rural-Marine	1.5	1.6
South Bend, Pa.	Semi-Rural	1.5	1.5
Kure Beach, N.C., 800-ft site	Marine (800 ft from ocean)	1.8	1.7
Point Reyes, Calif.	Marine	1.8	1.8
Sandy Hook, N.J.	Industrial-Marine	2.2	1.6
New York, N.Y. (spring exposure)	Industrial	3.1	3.6
Kearny, N.J.	Industrial	3.3	2.6
Halifax, N.S., Canada	Industrial-Marine ^b	3.8	18.0
New York, N.Y. (fall exposure)	Industrial	6.0	3.7
Daytona Beach, Fla.	Marine	7.1	2.6
Kure Beach, N.C., 80-ft site	Marine (80 ft from ocean)	13.0	5.7

^a While test site is 1500 ft. from brackish water, prevailing winds are from inland and prevent deposition of salt water spray.

^b Test site is near a smokestack; prevailing winds blow fumes over the test site.

Table 4-3.—THE CORROSION RATES OF METALS IN VARIOUS LOCATIONS.^a

Test Site	Corrosion rate, mil/year		
	Nickel 200	Monel alloy 400	Copper
Altoona, Pa. (Heavy industrial-R.R.)	0.222	0.076	0.055
New York, N.Y. (Urban-industrial)	0.144	0.062	0.054
State College, Pa. (Rural-farm)	0.0085	0.007	0.017
Phoenix, Ariz. (Rural-semiarid)	0.0015	0.002	0.005

$\mu\text{g}/\text{m}^3$ in New York City and $131 \mu\text{g}/\text{m}^3$ at Elizabeth, New Jersey, near Kearney,¹⁰ it should be pointed out that there are significant differences in levels of gaseous pollution, particularly sulfur dioxide, between State College and the larger industrial communities. Consequently, the differences in corrosion rates cannot be solely attributed to the effects of particulates.

In Chicago and St. Louis, steel panels were exposed at a number of sites, and measurements were taken of corrosion rates and of levels of sulfur dioxide and particulates.¹¹ In St. Louis, except for one exceptionally polluted site, corrosion losses correlated well with sulfur dioxide levels, averaging 30 percent to 80 percent higher than corrosion losses measured in nonurban locations. Over a 12-month period in Chicago, the corrosion rate at the most corrosive site (mean SO_2 level of 0.12 ppm) was about 50 percent higher than at the least corrosive site (mean SO_2 level of 0.03 ppm). Sulfation rates in St. Louis, measured by lead peroxide candle, also correlated well with weight loss due to corrosion. Although suspended particulate levels measured in Chicago with high-volume samplers correlated with corrosion rates, a covariance analysis indicated that sulfur dioxide concentrations had the dominant influence on corrosion. Measurements of dust-fall in St. Louis, however, did not correlate significantly with corrosion rates. Based on these data, it appears that considerable corrosion may take place (i.e., from 11 percent to 17 percent weight loss in steel panels) at annual average sulfur dioxide concentrations in the range of 0.03 ppm to 0.12 ppm, and although high particulate levels tend to accompany high sulfur dioxide levels, the sulfur dioxide concentration appears to have the more important influence.

Comparative studies of the rates of corrosion of zinc-coated steel panels exposed to various community atmospheres were conducted by Committee B-3 of the ASTM.⁹ The corrosion rates of zinc at four of the locations over a six-year period are shown in Figure 4-2, as weight loss versus time. For each location, the corrosion rate is essentially constant with time. The atmosphere of New York City with greater concentration

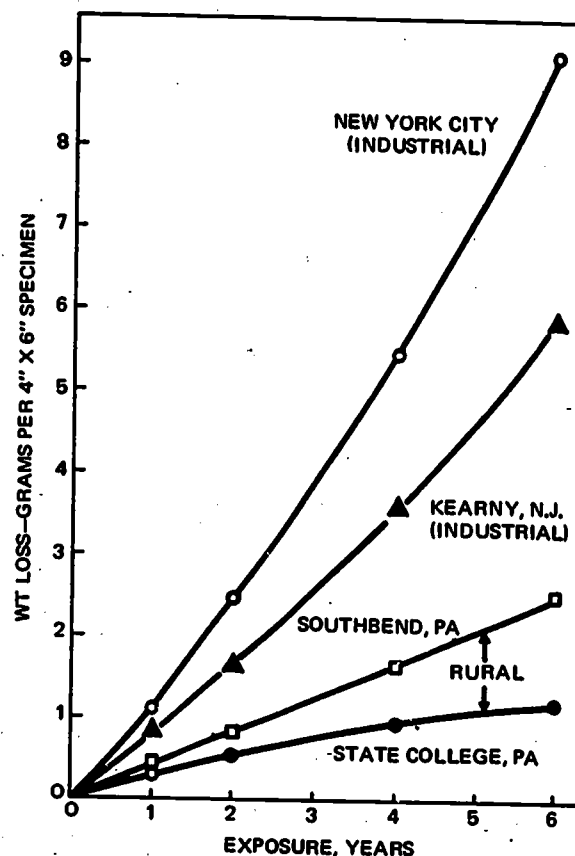


FIGURE 4-2. Weight Loss from Zinc Specimen as a Function of Exposure Time.⁹ (The figure shows the rate of corrosion of zinc at four locations in the United States, and indicates that corrosion is more rapid in industrial areas.)

of particles and sulfur oxides produces a steeper rate of corrosion of zinc than that of Kearny, New Jersey, which is also heavily polluted. Zinc corrodes at a greater rate in both industrial communities than in the rural or semirural sites in South Bend, Pennsylvania, and State College, Pennsylvania.⁹

Studies of the effects of air pollution on the atmospheric corrosion of three metals (nickel 200, Monel alloy 400, and copper) exposed to four diverse atmospheres (heavy-industrial, urban-industrial, rural-farm, and rural-semiarid) were conducted over a 20-year period.⁹ Nickel 200 is 99.5 percent nickel and Monel 400 is essentially 30 percent copper and 70 percent nickel. Though these metals are relatively corrosion-resistant, it will be noted from Table 4-3 that they are

corroded more in industrial atmospheres than in rural ones. However, because of the corrosion resistance of the metals, all of the rates are relatively low when compared to those for unalloyed iron or steel. In the industrial locations, nickel 200 corroded at rates two to four times greater than Monel (nickel-copper) alloy 400, while in the rural locations, its corrosion rate was about equal to that of Monel alloy 400, and only one-half the rate for copper. Even Monel 400, with its superior corrosion resistance, when used as a gutter in an industrial installation, became pitted when the soot was permitted to accumulate.⁹ Unburned carbon in the soot led to the formation of local galvanic corrosion cells. The result was premature perforation and accelerated attack of the metal sheet.

Larrabee¹² confirmed that the corrosion resistance of steel is greatly improved by the addition of small amounts of copper, or low alloys of chromium, nickel, copper, and phosphorus. The additional cost of such alloy steels when they are used to resist atmospheric corrosion is attributable to air pollution.

Particles can cause corrosion of electronic gear of all types even where precious metals are used to minimize such corrosion. Electrical instruments and electronic components are factors of growing importance in the computer and missile industries, and also monitor and control an increasingly large share of manufacturing processes.¹³ Oily or tarry particles, commonly found in industrial and urban areas, are serious factors in the corrosion and failure of electric contacts, connectors, and components.¹³

Dust can act mechanically or chemically on electric contacts. It can deposit on the surfaces and interfere with electrical contact closure, it can become imbedded in the surfaces of contacts, or it can induce wear by abrasion if the contacts slide.^{13,14} Hygroscopic dusts, accumulated on contacts, will absorb water to form thin electrolytic films which are corrosive to base metals.⁵ If the contact members are not of identical composition, galvanic corrosion may occur. The

terminal and corrosion films increase corrosion, avoided or minimized only by fabricating the electrical contacts out of nonreactive metals, by encapsulation, or by air purification devices such as filters or by gas-absorbing chemicals. Any of these solutions to the problem increases costs.

C. EFFECTS OF PARTICULATE MATTER ON BUILDING MATERIALS

Building materials and surfaces are soiled, disfigured, and damaged by atmospheric particles. Some of these stick to surfaces of stone, brick, paint, glass, and composition materials, forming a film of tarry soot and grit which may or may not be removed by the action of the rain. The result is a dingy, soiled appearance (Figure 4-3), a loss in aesthetic attractiveness and, in many cases, a physical-chemical degradation or erosion of these surfaces.

In cities where large quantities of soot-producing fuel are burned, the problem is particularly severe. Much money and effort have been spent on sandblasting off the sooty layers which have accumulated on prominent buildings in burning soot-producing fuel cities.¹⁵⁻¹⁷ The tarry substances or carbonaceous material resulting from inefficient combustion of soot-producing fuel are likely to be sticky and also acidic;² if not flushed off by rain they will adhere to surfaces and corrode them over extended periods of time.¹⁸ Smoke particles may also act as a reservoir for adsorbed gases, such as SO₂, and harmful acids, including sulfurous, sulfuric, and hydrochloric acids, as well as hydrogen sulfide. These materials cause the deterioration of many of the less resistant types of masonry.^{19,20}

Buildings in polluted areas become unsightly quickly and require periodic cleaning and maintenance to remove the tarry asphalt-like deposits. In Washington D.C. it was found necessary to clean the smoke and grime from the new Supreme Court building even before initial occupancy (in the mid-thirties).¹⁶ While the soiling effect of soot is by far the most evident to the eye, it does not in itself cause the deterioration of building material. The destruction is due to acids, alone or in combination with other pollutants.

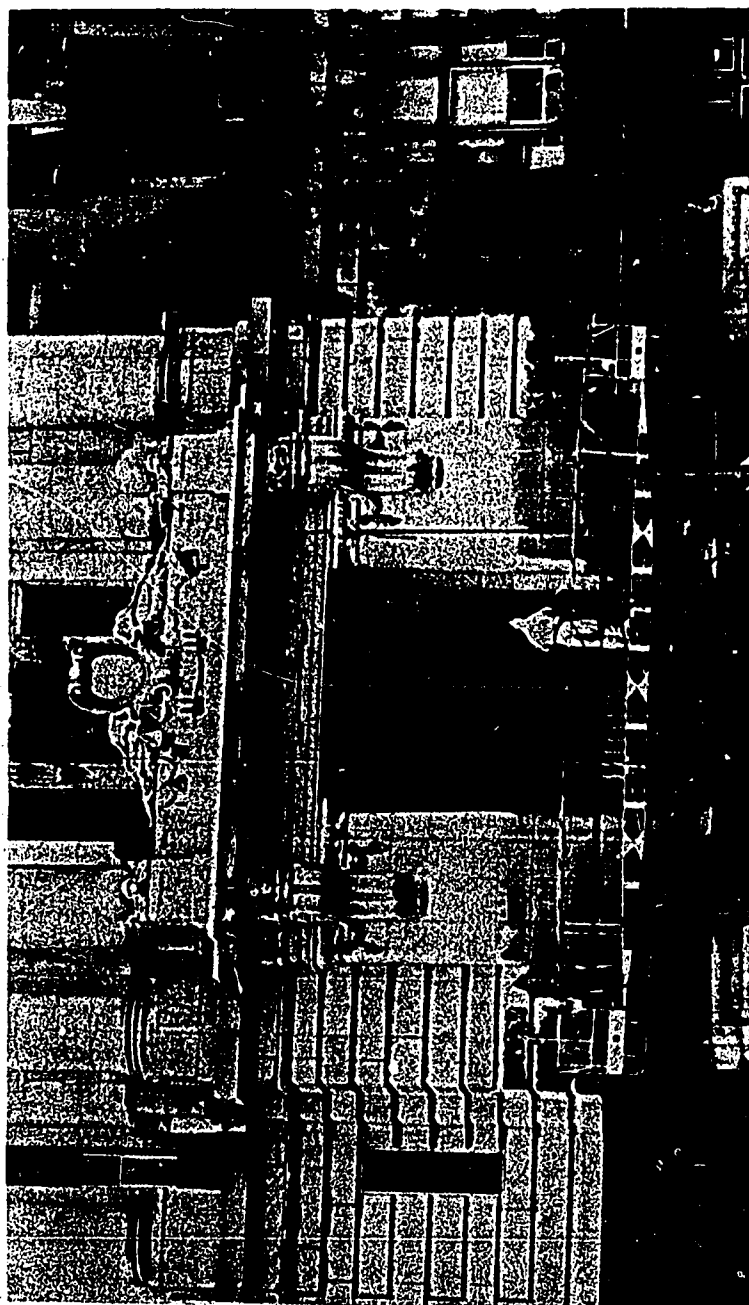


FIGURE 4-3. District Building, Washington, D.C. (This photograph shows sooty layers deposited on a public building, and the effect of sandblasting on restoring appearance. The photograph was taken on October 26, 1960.)

D. SOILING AND DETERIORATION OF PAINTED SURFACES

Painted surfaces, walls, and ceilings of homes and other buildings are soiled by tarry and other particulate substances in the atmosphere. Furthermore, there are both liquid and solid particles present in polluted air in the form of fumes and mists of varying chemical composition which react with painted surfaces.^{18,21} The damage done is both aesthetic and material, and may affect not only buildings but also paintings and other works of art.

Auto finishes have been observed to become pitted and stained by iron particles deposited on them from a metal grinding operation nearby,^{6,22} while chromic acid mist from an electroplating operation formed a brown stain on light colored cars and a "blushing" on darker shades of paint. Repainting was required because the color changes were not reversible by washing or polishing.^{18,23} Cars parked near demolition operations of brick buildings have been severely damaged by alkali mortar dust in the presence of moisture.¹⁸ Characteristic pitting of painted surfaces is also caused by sodium carbonate particles from such industrial processing as soda ash manufacture.

Water soluble chlorides and sulfates—mostly the iron, copper, calcium, and zinc salts—are commonly found in particulate samples from cities^{21,23,24} and in rainwater.²³ These water-soluble particles are potential sources of osmotic blistering of painted surfaces. The effects on weathering by small quantities of such particles have been examined²¹ during laboratory studies of accelerated aging of various paint panels. The presence of 0.1 ppm of iron in the water in the accelerated weather apparatus produced yellow staining, while 0.5 ppm of copper produced severe brown staining. These same effects may be expected on natural exposure.²¹

Dust particles settling on wet varnish and paint films produce visible imperfections and reduction in the electrical resistance and

protective coatings, two sets of metal panels were coated with a varnish formulation under a "dust-free" apparatus, and then one set was removed from the apparatus so that the varnish films dried in contact with the laboratory air. The panels were then immersed in 3 percent aqueous sodium chloride for seven and one-half months. Where the varnish film had dried in "normal" laboratory room air, considerable rusting occurred on the panels after six months of immersion; no corrosion occurred on the panels dried in the "dust-free" environment. Clearly the dust particles that settled on the films of the panels during the drying period were a predominant factor in initiating corrosion of the metal.²⁵

The exteriors of buildings in industrial areas, when repainted, collect numerous black specks on their surfaces even before the paint has dried.²⁶ Within two to four years depending on the degree of particle concentration, these building exteriors are distinctly soiled and require repainting.^{18,26}

E. SOILING AND DEGRADATION OF TEXTILES

Soiling of clothing, curtains, and other textiles not only diminishes their aesthetic appeal but also reduces their functional effectiveness. A garment which soils readily will not have the same user appeal as one which does not, even though its performance may be equal in other respects. Economic costs are therefore involved both in the extra cleaning of garments which soil readily and in the development and manufacture of soil-resistant textiles.

The extent of soiling of textiles, such as curtains, is influenced by various external factors such as temperature, relative humidity, and wind speed and the specific size and characteristics of the atmospheric particles. In addition, the degree of soiling is dependent on the construction and finish of the textile material and the type of fiber from which it is made.²⁷

When large particles are deposited on the surface of the textile, the soiling may be

as air sweeps through the intricate channels between them, i.e., the material behaves as a filter. The soiling of curtains and flags is an illustration of this effect. Airborne particles can also affect the cleanliness of yarns and fabrics during their manufacture. Economic loss to textile manufacturers can occur unless suitable air filters are used in manufacturing plants.²⁷ Laboratory studies by Rees,²⁷ using a dust circulating apparatus under controlled conditions; show that a closely woven fabric of low porosity best resists soiling by airborne particles.

If an exposed textile material acquires an electrostatic charge, for example by friction, and is able to retain its charge, charged particles of opposite sign will be attracted to it, thereby increasing its rate of soiling.²⁷ Cellulose acetate and some of the synthetic textiles acquire, by friction during spinning and weaving, electrostatic charges which, because of the high insulating properties of these materials, are retained for a long period of time. This results in troublesome "fog-marking," caused by attraction of airborne particles to the charged textile.²⁷

In laboratory investigations of the electrostatic attraction of airborne particles to cotton textiles, Rees²⁷ found that soiling is more rapid when the fabric is charged than when uncharged, and that the rate of soiling is increased by raising the applied electric potential (which increases the charge density on the fabric). For any given potential, the rate of soiling appears to be greater for a positively charged fabric specimen than for a negatively charged one.

The vulnerability of textile fabrics and furnishings to the acid components of particulate matter depends on the chemical composition of the textiles.²⁸ Cellulosic fibers, such as cotton, linen, hemp, jute, and man-made rayon are particularly sensitive to attack from such substances as sulfurous and sulfuric acids, while animal fibers, such as wool and furs, are more resistant to acid damage.²⁸

Curtains are particularly vulnerable to air pollutants and often deteriorate quickly hanging at open windows; soiling occurs to some extent even when windows are closed.

filter for acid-laden dust and soot. Airborne metallic iron and zinc particles, constituents of city dust,^{28, 29} are catalysts and promote oxidation of sulfur dioxide to sulfuric acid, which may contribute to textile degradation. Curtains weakened by conditions of exposure arising partly from atmospheric soiling and acidity give way in a characteristic manner by splitting in parallel lines along the folds where the greatest number of particles accumulated.

F. SUMMARY

Airborne particles—including soot, dust, fumes, and mist—can, according to their chemical composition and physical state, cause a wide range of damage to materials. They may cause deterioration merely by settling on surfaces and soiling them thus creating a need for more frequent cleaning which in itself weakens materials. More importantly, they can cause direct chemical damage to materials in two ways: First, through their own intrinsic corrosiveness, and secondly, through the action of corrosive chemicals which they may have absorbed or adsorbed. Airborne particulates have been implicated in the corrosion of metals and metallic surfaces; in the soiling, damage, and erosion of buildings and other structures; in the discoloration and destruction of painted surfaces; and in the aesthetic degradation and damage of fabrics and clothing.

Metals ordinarily can resist corrosion in dry air alone, or even in moist clean air. Even inert dust or soot, in the absence of active chemical agents, has little effect on metal surfaces. However, hygroscopic particles commonly found in the atmosphere can corrode metal surfaces although no other pollutants may be present. This was shown in laboratory studies in which steel test panels were dusted with various common particulates. Although corrosion rates were low at relative humidities under 70 percent, they tended to increase with increased humidity. The addition of sulfur dioxide to the laboratory air greatly accelerated the rates of corrosion.

In general, there is an increasing rate of

environments, the least polluted areas, to heavily polluted urban industrial sites. Samples of iron, for example, have shown weight loss due to corrosion that is four times greater in industrial atmospheres than in rural atmospheres. Steel samples corroded in one year 3.1 times as much in New York City (spring exposure), where the particulate concentration was $176 \mu\text{g}/\text{m}^3$, as in the rural atmosphere of State College, Pennsylvania, where the mean concentration could be expected to range about $60 \mu\text{g}/\text{m}^3$ to $65 \mu\text{g}/\text{m}^3$. Steel samples exposed in New York in the fall of the year, when the particulate and sulfur dioxide levels are higher than in the spring, corroded 6 times faster than the samples at State College. Similarly, zinc samples exposed in New York corroded about 3.6 times as much as those at State College, while zinc samples at Kearny, New Jersey, corroded about 2.6 times as much (the concentration was $131 \mu\text{g}/\text{m}^3$ at Elizabeth, New Jersey, near Kearney).

It should be concluded, however, that the variation in corrosion rates referred to above are due only to differences in particulate matter, since there are significant differences in gaseous pollutant concentrations between State College and the other areas. Even highly resistant metals have shown corrosion rates which are progressively larger over the following range of environments; (1) a rural semiarid site; (2) a rural farm environment; (3) an urban industrial area; (4) a heavy industrial area. In Chicago and St. Louis, steel panels were exposed at a number of sites, and measurements were taken of corrosion rates and of levels of sulfur dioxide and particulates. In St. Louis, except for one exceptionally polluted site, corrosion losses correlated well with sulfur dioxide levels, averaging 30 percent to 80 percent higher than corrosion losses measured in nonurban locations. Over a 12-month period in Chicago, the corrosion rate at the most corrosive site (mean SO_2 level of 0.12 ppm) was about 50 percent higher than at the least corrosive site (mean SO_2 level of 0.03 ppm). Sulfation rates in St. Louis, measured by lead peroxide candle

levels measured in Chicago with high-volume samplers correlated with corrosion rates, a covariance analysis indicated that sulfur dioxide concentrations had the dominant influence on corrosion. Measurements of dustfall in St. Louis, however, did not correlate significantly with corrosion rates. Based on these data, it appears that considerable corrosion may take place (i.e., from 11 percent to 17 percent weight loss in steel panels) at annual average sulfur dioxide concentrations in the range of 0.03 ppm to 0.12 ppm, and although high particulate levels tend to accompany high sulfur dioxide levels, the sulfur dioxide concentration appears to have the more important influence.

Particles play a significant role in corrosion and damage to electronic equipment of all kinds, even when precious metals are used to minimize such effects. The contacts in electrical switches are vulnerable to chemical or mechanical action by particulates. The particulates commonly found in industrial and urban atmospheres are serious factors in the corrosion and failure of electrical connectors and circuits.

The ability of particulates to damage and soil buildings, sculpture, and other structures is particularly great in cities where large quantities of coal and sulfur-bearing fuel oil are burned. Particles may cause deterioration to many types of masonry by acting as reservoirs for the harmful acids generated by the combustion of these fuels. In addition to direct erosion and physical degradation of materials, particles stick to surfaces with which they come in contact, forming a film of tarry soot and grit which may or may not be removed by the action of rain. The result is a dingy, soiled appearance, and much money and effort must be spent to sandblast off the sooty layers that accumulate.

Particles also may soil the painted surfaces of walls, ceilings, and the exteriors of homes and buildings, and, under certain circumstances, may cause them to become stained and pitted. Automobile finishes, for example, have been damaged by particulate matter emitted from nearby industrial on-

zinc salts—are commonly found in particulate samples from cities and in rainwater and may cause blistering and enhance the weathering of painted surfaces. Particles may settle on painted surfaces before the paint has dried, thus producing imperfections and reducing the ability of the paint to protect the surface. Such surfaces are likely to soon require refinishing.

The soiling of clothing, curtains, and similar textiles makes them unattractive and diminishes their use. The extent of soiling is influenced by a number of factors, including the temperature, the relative humidity, and the wind conditions. Small particles may penetrate deep into the fibers of curtains hanging in open windows. Curtains weakened by such exposure to atmospheric soiling and acidity deteriorate in a characteristic manner. The vulnerability of textile products to the acid components of airborne particles also depends on the composition of the material. Cellulosic fibers, for example, such as cotton, linen, hemp, jute, and man-made rayon, are particularly sensitive to attack from such substances as sulfurous and sulfuric acids. In addition to the aesthetic degradation and the nuisance created by particulate matter, direct costs may be associated with the increased rate of deterioration of textiles, the extra cleaning required, and the manufacturing expenses for fabrics that are more resistant to air pollution.

G. REFERENCES

1. Sheleikhovskii, G. V. "Smoke Pollution of Towns." *Akademiya Kommunal'nogo Khzyaistva im. K. D. Pamfilova. Academy of Municipal Economy im. K. D. Pamfilova. Izdatel'stvo Ministerstva Kommunal'nogo Khozyaistva RSFSR, Moskva-Leningrad, 1949.* (Translated from Russian and Published for the National Science Foundation, Washington, D.C. by the Israel Program for Scientific Translations, Jerusalem, 1961.)
2. Greenburg, L. and Jacobs, M. B. "Corrosion Aspects of Air Pollution." *Am. Paint J.*, Vol. 39, pp. 64-78, 1955.
3. Vernon, W. H. J. "The Corrosion of Metals." *J. Roy. Soc. Arts*, Vol. 97, pp. 578-610, 1949.
4. Vernon, W. H. J. "A Laboratory Study of the Corrosion of Metals by Air Pollution." *Trans. Faraday Soc.*, Vol. 31, pp. 1668-1700, 1935.
5. Preston, R. St. J. and Sanyal, B. "Atmospheric Corrosion by Nuclei." *J. Appl. Chem.*, Vol. 6, pp. 28-44, 1956.
6. Tice, E. A. "Effects of Air Pollution on the Atmospheric Corrosion Behavior of Some Metals and Alloys." *J. Air Pollution Control Assoc.*, Vol. 12, pp. 533-559, 1962.
7. Hudson, J. D. "Present Position of the Corrosion Committee's Field Tests on Atmospheric Corrosion (Unpainted Specimens)." *J. Iron Steel Inst.*, Vol. 148, pp. 161-215, 1943.
8. Larrabee, C. P. and Ellis, O. B. "Report of a Subgroup of Subcommittee VII on Corrosiveness of Various Atmospheric Test Sites as Measured by Specimens of Steel and Zinc. Committee B-3, American Society for Testing Materials." *Am. Soc. Testing Mater. Proc.*, Vol. 59, pp. 183-201, 1959.
9. Copson, H. R. "Report of Subcommittee VI. Committee B-3, American Society for Testing Materials." In: *Corrosion of Nonferrous Metals Symposium*, Am. Soc. Testing Mater., Spec. Tech. Pub. 175, 1955.
10. "Air Pollution Measurements of the National Air Sampling Network (Analysis of Suspended Particulates, 1957-1961)." U.S. Dept. of Health, Education, and Welfare, Div. of Air Pollution, Washington, D.C., 1962.
11. Upham, J. B. "Atmospheric Corrosion Studies in Two Metropolitan Areas." *J. Air Pollution Control Assoc.*, Vol. 17, pp. 398-402, 1967.
12. Larrabee, C. P. "Effect of Composition and Environment on Corrosion of Iron and Steel." *Am. Soc. Metals*, pp. 30-50, 1956.
13. Antler, M. and Gilbert, J. "The Effects of Air Pollution on Electric Contacts." *J. Air Pollution Control Assoc.*, Vol. 13, pp. 405-415, 1963.
14. Williamson, J. B. P., Greenwood, J. A. and Harris, J. "The Influence of Dust Particles on the Contact of Solids." *Proc. Roy. Soc., Ser. A (London)*, Vol. 237, pp. 560-573, 1956.
15. Carey, W. F. "Atmospheric Deposits in Britain: A Study in Dinginess." *Intern. J. Air Pollution*, Vol. 2, pp. 1-26, 1956.
16. Meller, H. B. and Sisson, L. B. "Effects of Smoke on Building Materials." *Ind. Eng. Chem.*, Vol. 27, pp. 1309-1312, 1935.
17. Le Clerc, E. "Economic and Social Aspects of Air Pollution." In: *Air Pollution (World Health Organization)*, Columbia University Press, New York, 1961, pp. 279-291.
18. Yocom, J. E. "The Deterioration of Materials in Polluted Atmospheres." *J. Air Pollution Control Assoc.*, Vol. 8, pp. 203-208, 1958.
19. Cartwright, J. Nagelschmidt, G., and Skidmore, J. W. "The Study of Air Pollution with the

20. Giles, C. H. "Adsorption. II. In Air Pollution and in Natural Processes." Chem. Ind. (London), Vol. 19, pp. 770-781, 1964.
21. Holbrow, G. L. "Atmospheric Pollution: Its Measurement and Some Effects on Paint." J. Oil Colour Chemists Assoc., Vol. 45, pp. 701-718, 1962.
22. Fochtman, E. C. and Langer, G. "Automobile Paint Damaged by Airborne Iron Particles." J. Air Pollution Control Assoc., Vol. 6, pp. 243-247, 1957.
23. Tabor, E. C. and Warren, W. V. "Distribution of Certain Metals in the Atmosphere of Some American Cities." Arch. Ind. Health, Vol. 17, pp. 145-151, 1958.
24. Waller R. E., Brooke, A. G. F., and Cartwright, J. "An Electron Microscope Study of Particles in Town Air." Intern. J. Air Water Pollution, Vol. 7, pp. 779-786, 1963.
25. Graff-Baker, C. "The Effect of Dust Particles on the Electrical Resistance and Anti-Corrosive Properties of Varnish and Paint Films." J. Appl. Chem., Vol. 8, pp. 500-598, 1958.
26. Parker, A. "The Destructive Effects of Air Pollution on Materials." The Sixth Des Voeux Memorial Lecture. (Presented at the Annual Conference of the National Smoke Abatement Society, Bournemouth, England, September 28, 1955.)
27. Rees, W. H. "Atmospheric Pollution and Soiling of Textile Materials." Brit. J. Appl. Phys., Vol. 9, pp. 301-305, 1958.
28. Petrie, T. C. "Smoke and the Curtains." Smokeless Air, Vol. 18, pp. 62-64, 1948.
29. Tebbens, B. D. "Residual Pollution Products in the Atmosphere." In: Air Pollution, Chapter 2, Vol. 1, 1st edition, A. C. Stern (ed.), Academic Press, New York, 1962, pp. 23-40.

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